

Activity No. 3: Install SSH server on CentOS or RHEL 8

Course Code: CPE 212

Course Title: Automating Server Management

Intended Learning Outcomes Addressed

At the end of the activity the students will be able to:

- Install Community Enterprise OS or Red Hat Linux OS
- Configure remote SSH connection from remote computer to CentOS/RHEL-8

Objectives

This laboratory guides the students to Install Community Enterprise OS and set up SSH key-based authentication between local and remote machines.

1. DISCUSSION

CentOS vs. Debian: Overview

CentOS and Debian are Linux distributions that spawn from opposite ends of the candle.

CentOS is a free downstream rebuild of the commercial Red Hat Enterprise Linux distribution where, in contrast, Debian is the free upstream distribution that is the base for other distributions, including the Ubuntu Linux distribution.

As with many Linux distributions, CentOS and Debian are generally more alike than different; it isn't until we dig a little deeper that we find where they branch.

CentOS vs. Debian: Architecture

The available supported architectures can be the determining factor as to whether a distro is a viable option or not. Debian and CentOS are both very popular for x86_64/AMD64, but what other archs are supported by each?

Both Debian and CentOS support AArch64/ARM64, armhf/armhfp , i386 , ppc64el/ppc64le. (Note: armhf/armhfp and i386 are supported in CentOS 7 only.)

CentOS 7 additionally supports POWER9 while Debian and CentOS 8 do not. CentOS 7 focuses on the x86_64/AMD64 architecture with the other archs released through the AltArch SIG (Alternate Architecture Special Interest Group) with CentOS 8 supporting x86_64/AMD64, AArch64 and ppc64le equally.

Debian supports MIPSel, MIPS64el and s390x while CentOS does not. Much like CentOS 8, Debian does not favor one arch over another—all supported architectures are supported equally.

CentOS vs. Debian: Package Management

Most Linux distributions have some form of package manager nowadays, with some more complex and feature-rich than others.

CentOS uses the RPM package format and YUM/DNF as the package manager.

Debian uses the DEB package format and dpkg/APT as the package manager.

Both offer full-feature package management with network-based repository support, dependency checking and resolution, etc.. If you're familiar with one but not the other, you may have a little trouble switching over, but they're not overwhelmingly different. They both have similar features, just available through a different interface.

2. MATERIALS AND EQUIPMENT

- VirtualBox
- CentOS 9 Stream
- Personal Computer

3. PROCEDURE

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS 9 Stream here: <https://www.centos.org/download/>

CentOS Stream

Continuously delivered distro that tracks just ahead of Red Hat Enterprise Linux (RHEL) development, positioned as a midstream between Fedora Linux and RHEL. For anyone interested in participating and collaborating in the RHEL ecosystem, CentOS Stream is your reliable platform for innovation.

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Architecture	ISOs	RPMS	Cloud	Containers	Vagrant
x86_64	Mirrors	Mirrors	Images	Images	Boxes
ARM64 (aarch64)	Mirrors	Mirrors	Images	Images	
IBM Power (ppc64le)	Mirrors	Mirrors	Images	Images	
IBM Z (s390x)	Mirrors	Mirrors	Images	Images	

Documentation

[Release Announcement](#)
[Release Notes](#)

End-of-life

2027-05-31 (End of [RHEL 9 full support](#) phase)

2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.

Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
`$ dnf install openssh-server`
2. Start the *sshd* daemon and set to start after reboot:
`$ systemctl start sshd`
`$ systemctl enable sshd`
3. Confirm that the sshd daemon is up and running:
`$ systemctl status sshd`
4. Open the SSH port 22 to allow incoming traffic:
`$ firewall-cmd --zone=public --permanent --add-service=ssh`

\$ firewall-cmd --reload

5. Locate the ssh server man config file */etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the */etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:

\$ systemctl reload sshd

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.
3. On CentOS, verify that you have the *authorized_keys*.

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

4. SUPPLEMENTARY ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
2. What are the main differences between Debian and Red Hat Linux distributions?

Conclusions/Learnings:

5. REFERENCES